

Lecture #4

Content to be covered

- ☐ Colours in thin films
- ☐ Anti-reflection films
- ☐ Wedge shaped films

Resources to be consulted

- ☐ OPTICS by Ajay Ghatak

Colors in thin films

- The incident light is split up by reflection at the top and bottom of the film. The split rays are in a position to interfere and interference of these rays is responsible for colors.
- Since the interference condition is a function of thickness of the film, the wavelength and the angle of refraction, different colors are observed at different positions of the eye.
- It should be noted here that the conditions for maxima and minima in transmitted light are opposite to that of reflected light. Hence, the colors that are absent in reflected light will be present in transmitted light. The colors observed in transmitted light and reflected light are complimentary.
- An excessively thin film appears black in reflected system when illuminated with an extended source of white light.

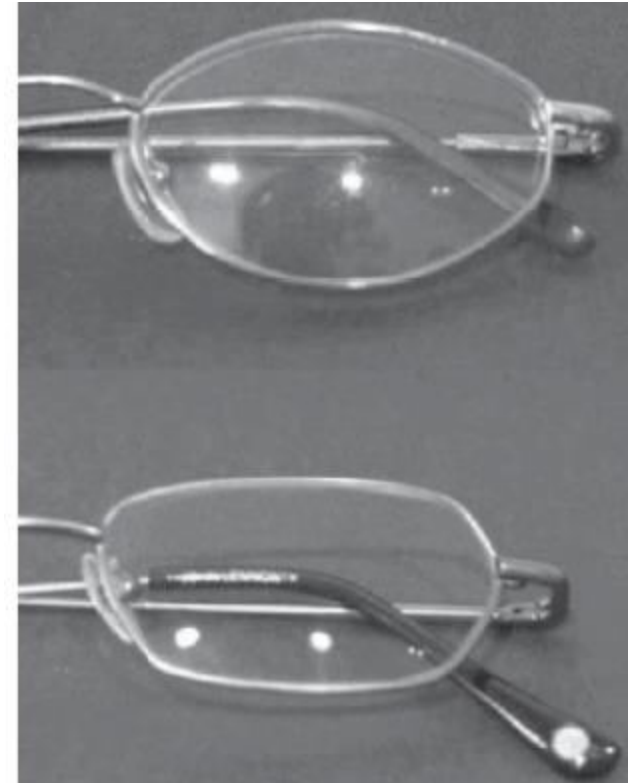
Non-reflecting Films

Figure shows a comparison between a glasses lens without anti-reflective coating (top) and a lens with anti-reflective coating (bottom). Note the reflection of the photographer in the top lens and the tinted reflection in the bottom.

The loss of energy due to reflection is one major reason of clarity reduction, meaning hereby, reduction in the intensity of the images since less light is transmitted through the lenses.

When films are coated on surface of lens, the reflectivity of these surfaces is appreciably reduced . Example CaF_2 , MgF_2

No light is destroyed by non reflecting film, but there is redistribution means decrease in reflection results an increase in transmission.



Non-reflecting Films

One of the important application of the thin film interference phenomenon lies in **reducing the reflectivity** of lens surfaces.

Now we will use the thin film interference phenomenon in reducing the reflectivity of lens surfaces. We all know that in many optical instruments such as telescope, there are many interfaces, and the loss of intensity due to reflections can be severe. For example, for near normal incidence, the **reflectivity of the crown glass surface in air is 4%.**

$$\left[\frac{\mu - 1}{\mu + 1} \right]^2 = \left[\frac{1.5 - 1}{1.5 + 1} \right]^2 \approx 0.04$$

In order to reduce these losses, lens surfaces are often coated with a **$\lambda/4\mu$** thick non-reflecting film. The refractive index of the film should be less than that of lens. The film thickness should be such that

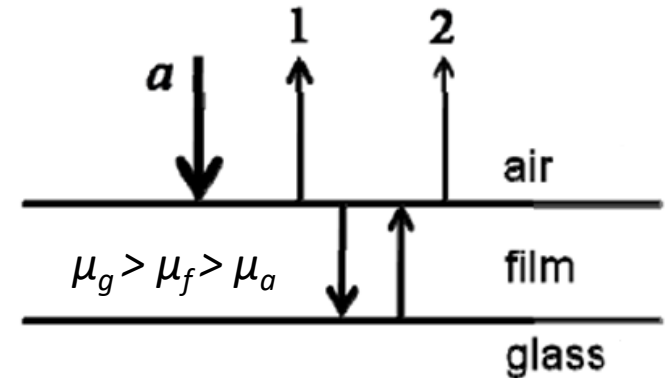
$$2\mu t = \frac{\lambda}{2}$$

Non-reflecting Films

$$\mu_a (\text{air}) = 1$$

$$\mu_f (\text{MgF}_2) = 1.38$$

$$\mu_g (\text{glass}) = 1.5$$



For normal incidence, the two interfering beams will interfere destructively, if

$$2 \mu_f t \cos r = (2n+1)\lambda/2$$

So, for minimum thickness, $n = 0$, $2 \mu_f t = \lambda/2 \Rightarrow t = \lambda/4\mu_f$

If a film, having thickness of $\lambda/4\mu_f$ and refractive index (μ_f) less than that of the glass (μ_g), is coated on glass, then waves reflected from the upper surface of the film interfere destructively with the waves reflected from the lower surface of the film, leading to **destructive interference**. Such films are known as a non or anti-reflecting films with refractive index μ_f is such that $\mu_a < \mu_f < \mu_g$.

Non-reflecting Films

$$r_1 = \frac{\mu_a - \mu_f}{\mu_a + \mu_f}, t_1 = \frac{2\mu_a}{\mu_a + \mu_f}$$

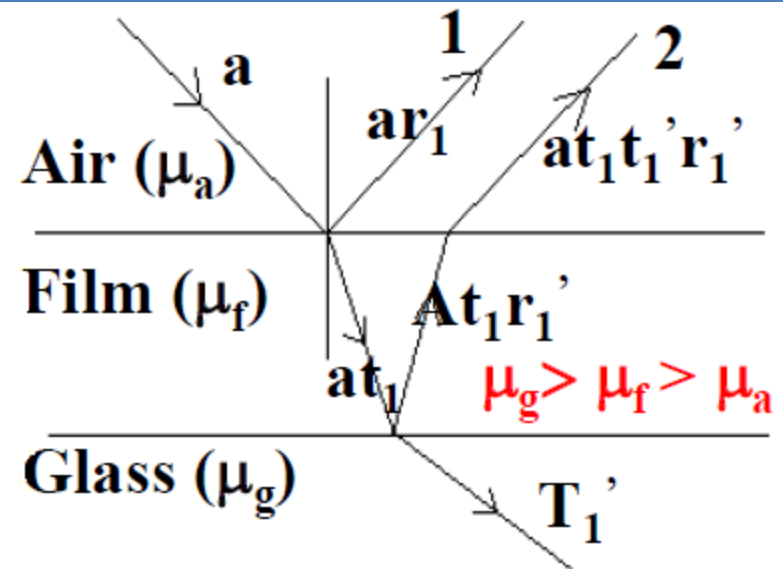
$$r_1' = \frac{\mu_f - \mu_g}{\mu_f + \mu_g}, t_1' = \frac{2\mu_f}{\mu_f + \mu_g}$$

Amplitude of ray 1: $ar_1 = \frac{\mu_a - \mu_f}{\mu_a + \mu_f} a,$

Amplitude of ray 2:

$$\begin{aligned} ar_1' t_1' t_1' &= a \frac{\mu_f - \mu_g}{\mu_f + \mu_g} \times \frac{2\mu_a}{\mu_a + \mu_f} \times \frac{2\mu_f}{\mu_f + \mu_a} \\ &= a \frac{4\mu_a \mu_f}{(\mu_f + \mu_a)^2} \times \frac{\mu_f - \mu_g}{\mu_f + \mu_g} \end{aligned}$$

This equation gives the estimate of refractive index of film which should be coated on a glass surface to reduce its reflectivity. If $\mu_a = 1$ then $\mu_f = \sqrt{\mu_g}$



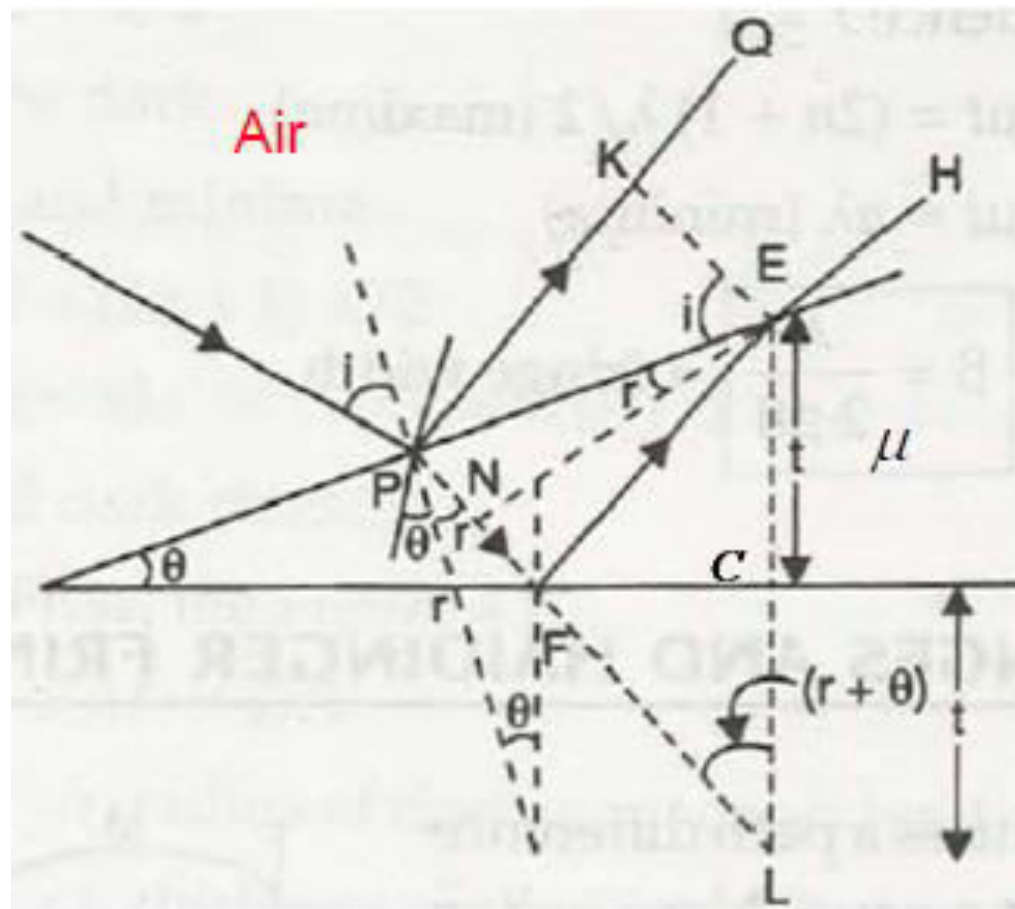
For complete destructive interference ray 1 and 2 must have the same amplitude, i.e. $ar_1 = ar_1' t_1' t_1'$

$$\frac{\mu_a - \mu_f}{\mu_a + \mu_f} = \frac{4\mu_a \mu_f}{(\mu_f + \mu_a)^2} \times \frac{\mu_f - \mu_g}{\mu_f + \mu_g}$$

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$$\mu_f = \sqrt{\mu_a \mu_g}$$

Interference In Wedge Shaped Films



The optical path difference between the rays PQ and EH is

$$\Delta = \mu(PF + FE) - PK$$

$$\Delta = \mu(PN + NF + FE) - PK$$

According to the Snell's law,

$$\mu = \frac{\sin i}{\sin r} = \frac{PK}{PN}$$

$$\Rightarrow PK = \mu PN$$

In the figure, $\triangle ECF$ and $\triangle FCL$ are congruent, thus $EC = CL = t$ and $FE = FL$

$$\Rightarrow \Delta = \mu(NF + FL) = \mu NL$$

$$\text{In } \triangle ENL, \quad \cos(r + \theta) = \frac{NL}{EL}$$

$$\Rightarrow NL = EL \cos(r + \theta) = 2t \cos(r + \theta)$$

$$\Rightarrow \Delta = 2\mu t \cos(r + \theta)$$

Interference In Wedge Shaped Films

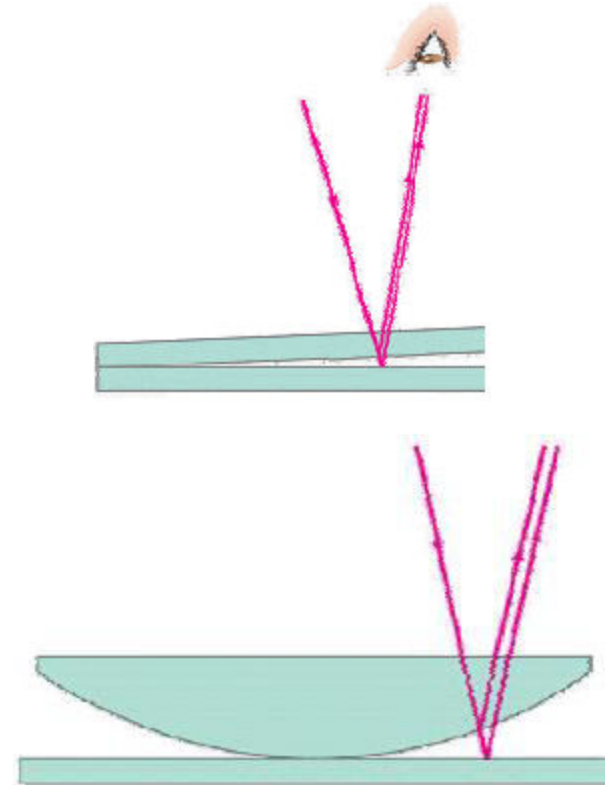
In practical applications, where θ is very small,

$$\cos(r + \theta) \approx \cos r$$

- Constructive interference $2\mu t \cos r = (2n + 1) \frac{\lambda}{2}$
- Destructive interference $2\mu t \cos r = n\lambda$

where $n = 0, 1, 2, \dots$

For monochromatic light beam incident on a wedge shaped film λ and μ are constant. So change in path difference is only due to **varying thickness of the film**. At a particular point, thickness is constant. So we get a bright or dark fringe at that point due to constant path difference.



Interference In Wedge Shaped Films

➤ **Fringe at Apex is Dark**

➤ **Straight and parallel fringes**

Locus of points having same thickness lie along lines parallel to the contact edge

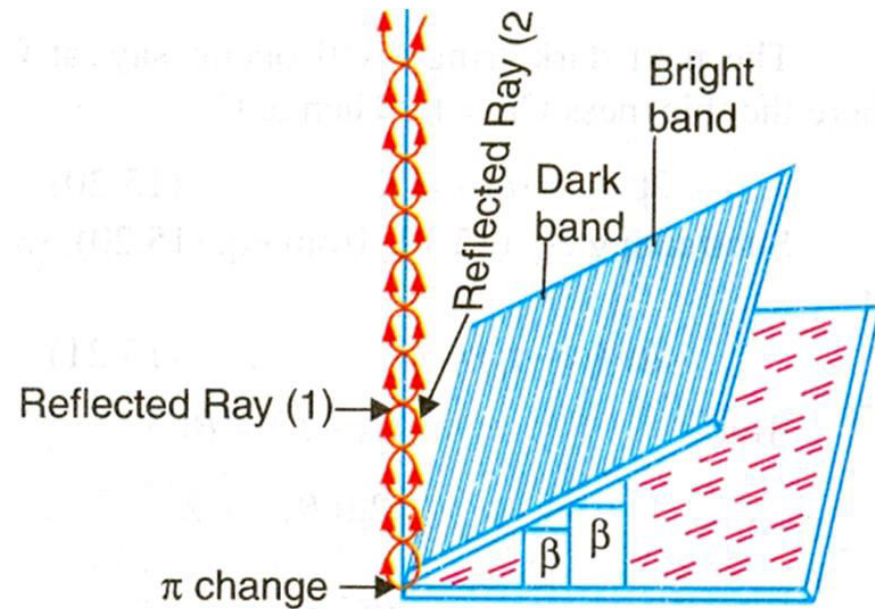
➤ **Equidistant fringes**

$$\beta \approx \frac{\lambda}{2\mu\theta}$$

➤ **Localized fringes**

Fringes formed close to the top surface of wedge.

➤ **Fringes of equal thickness**



Interference In Wedge Shaped Films

Determination of Wedge Angle

Using microscopes the positions of dark fringes at two distinct points Q and R , $OQ = x_1$, $OR = x_2$

Thickness of the wedge t_1 at Q , t_2 at R

For dark fringes at Q

$$\text{For small } \theta \quad 2\mu t_1 = m\lambda$$

$$\text{Therefore, } t_1 = x_1 \tan \theta \cong x_1 \theta$$

$$\text{For Dark fringe at } Q, \quad 2\mu x_1 \theta = m\lambda \quad \dots(1)$$

$$\text{For Dark fringe at } R, \quad 2\mu x_2 \theta = n\lambda$$

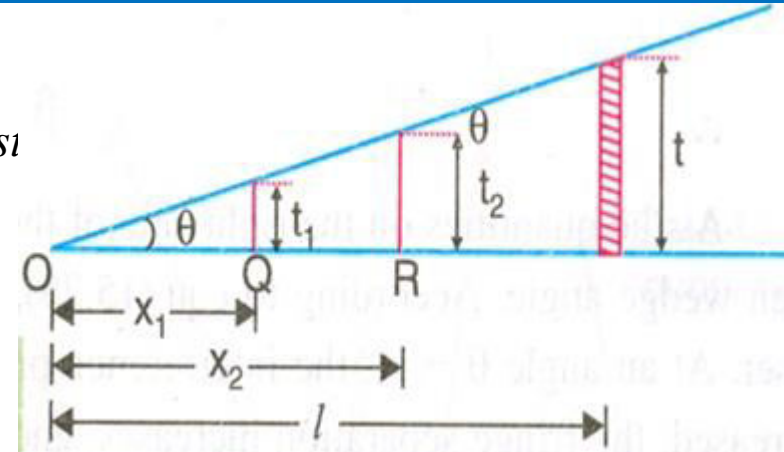
if N = number of dark fringes between Q and R .

$$\text{then } 2\mu x_2 \theta = (m + N)\lambda \quad \dots(2)$$

$$\text{from eq 1 and 2} \quad 2\mu(x_2 - x_1)\theta = N\lambda$$

Fringe width: if $N=1$ then $(x_2 - x_1) = \beta$

$$\beta = \frac{\lambda}{2\mu\theta}$$



$$\theta = \frac{N\lambda}{2\mu(x_2 - x_1)}$$

Interference In Wedge Shaped Films

The two glass surfaces are in contact at one end and separated at the other end by a thin wire. If the sodium light is incident normally, 20 interference fringes are observed between these edges. What is the thickness of the wire?

$$t = 10\lambda = 5.893 \times 10^{-6} \text{m}.$$

Explain what happens to the fringe width (i) if wedge angle is decreased and (ii) oil is introduced between surfaces of the glass plates in a wedge shaped films.

Interference In Wedge Shaped Films

- Fringes of equal thickness
- Fringe at the apex is dark.
- Fringes are straight, parallel and equidistant.
- Fringes are localized
- Fringes formed will be parallel to the edge OO' .
- bright and dark fringes of different orders will be observed at different thickness of the film.

